

Capstone Project: Problem Statement Assignment

Work together as a group to write a brief memo giving your problem statement. Your memo should answer the following questions (without directly quoting the wording below):

- What problem will your design team solve? (Or, what opportunity is your team trying to meet?) Clearly state the problem in a sentence or two, providing sufficient additional background so that someone outside the course could read and understand the motivation and need. Include (and cite) any relevant schematics, photos, or other figures to clarify the problem statement.
- Why is the problem important? What are potential impacts of a solution to this problem on society, industry, the environment, or others?
- What solutions are currently being used, if any?
- Who is your client and/or potential users?
- What are some necessary characteristics of a successful design? (These do not need to be as specific as design criteria yet; it is sufficient to provide preliminary characteristics or goals only here. If you do already have some design criteria from your client, include them here, along with any given functional requirements and/or engineering specifications.)
- What previous ME course work do you anticipate using to develop a design, conduct a thorough engineering analysis, and test your design prototypes?
- What ethical considerations do you anticipate needing to consider in your design decisions?
- What environmental and sustainability considerations do you anticipate needing to consider in your work?
- What safety concerns are evident at this preliminary stage of your design work?

Your memo should be in the form of a technical memorandum (written in third person) addressing the ME421 faculty and including your names (in the header). Any included links in the text should link to supplemental information only. After submitting this document, do continue using it (or the information in it), along with other shared documentation (i.e., meeting notes, working ideas, etc.), to keep records of your process.